Homework 4

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The air quality dataset

```
# load the data:
aqi_data <- load("C:/Users/FRT/Downloads/Fall-2026/MATH-8830/Dataset/pm.Rdata")
cat("The number of row in the data05", nrow(data05))
```

```
## The number of row in the data05 147341
```

```
cat("The number of row in the data14", nrow(data14))
```

```
## The number of row in the data14 373333
```

```
cat("The number of column in the data05", ncol(data05))
```

```
## The number of column in the data05 29
```

```
cat("The number of column in the data14", ncol(data14))
```

```
## The number of column in the data14 29
```

Question 1

```
#
library(tidyverse)
```

```
## Warning: le package 'tidyverse' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'ggplot2' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'tibble' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'tidyr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'readr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'purrr' a été compilé avec la version R 4.4.3

## Warning: le package 'dplyr' a été compilé avec la version R 4.4.3

## Warning: le package 'stringr' a été compilé avec la version R 4.4.3

## Warning: le package 'forcats' a été compilé avec la version R 4.4.3

## Warning: le package 'lubridate' a été compilé avec la version R 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

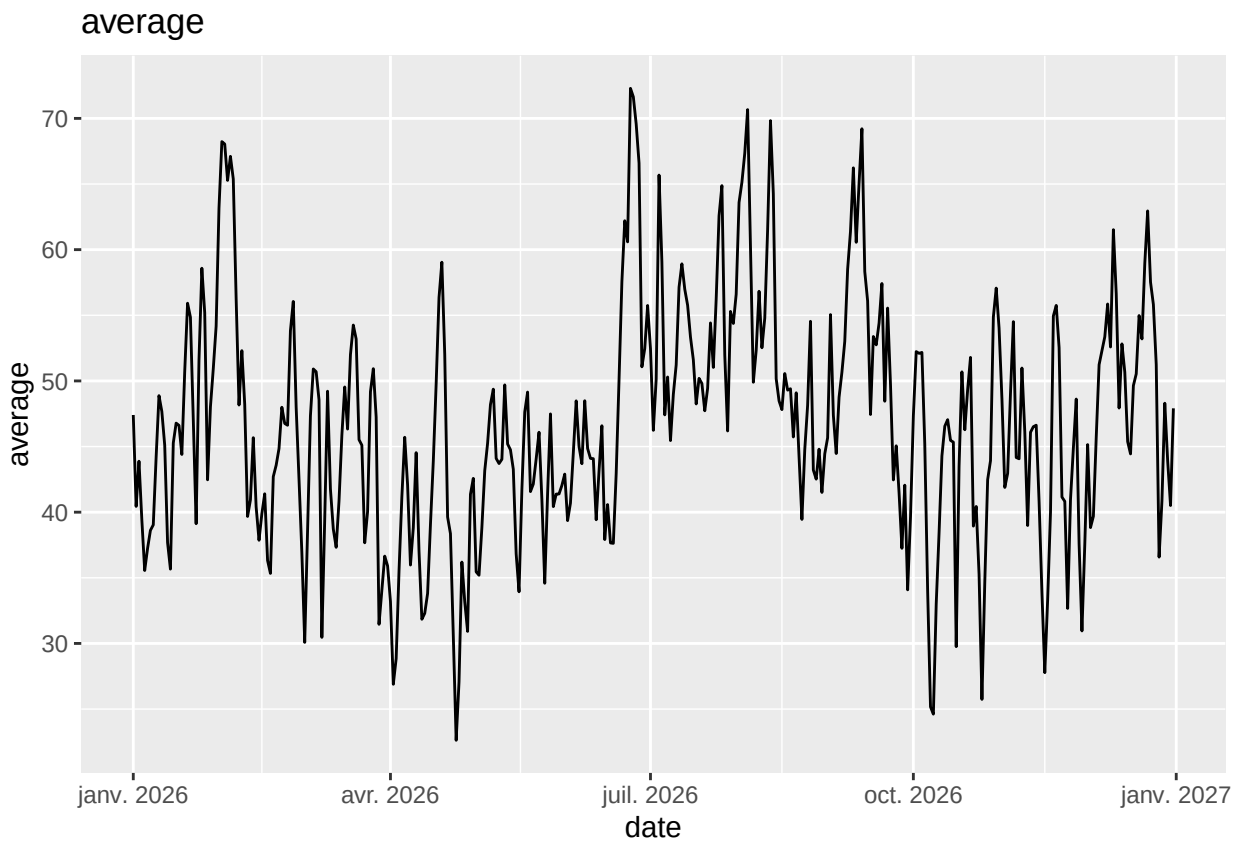
```
library(patchwork)
```

```
## Warning: le package 'patchwork' a été compilé avec la version R 4.4.3
```

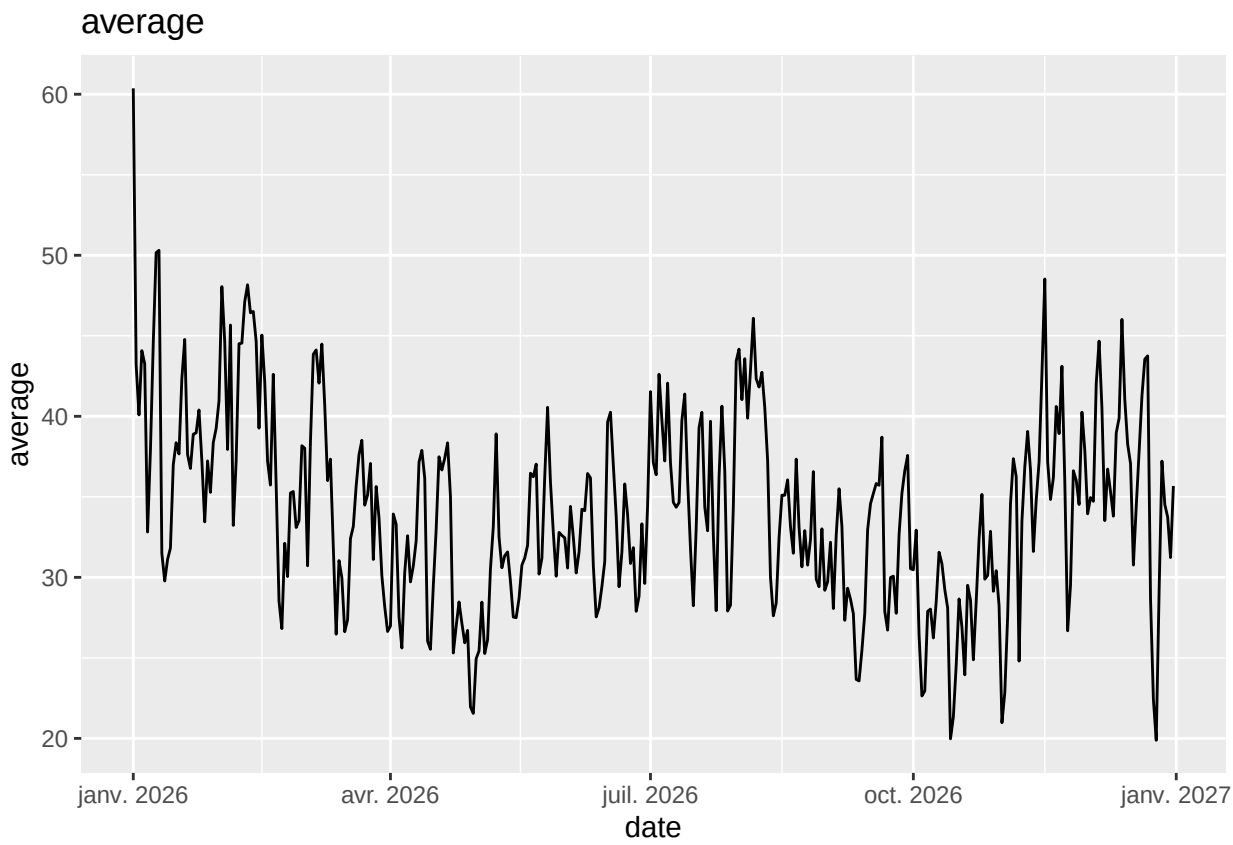
```
plot1 <- data05 |> group_by(Date.Local)|>
  summarize(average = mean(AQI, na.rm=TRUE),
            .groups = "drop_last") |>
   mutate(date = update(ymd(as.character(Date.Local)), year=2026))

plot2 <- data14 |> group_by(Date.Local) |>
  summarize(average = mean(AQI, na.rm=TRUE),
            .groups = "drop_last") |>
  mutate(date = update(ymd(as.character(Date.Local)), year=2026))

plot1 |>
  ggplot() +
  geom_line(aes(x=date, y=average)) +
  # scale_x_log10() +
  labs(title = "average")
```



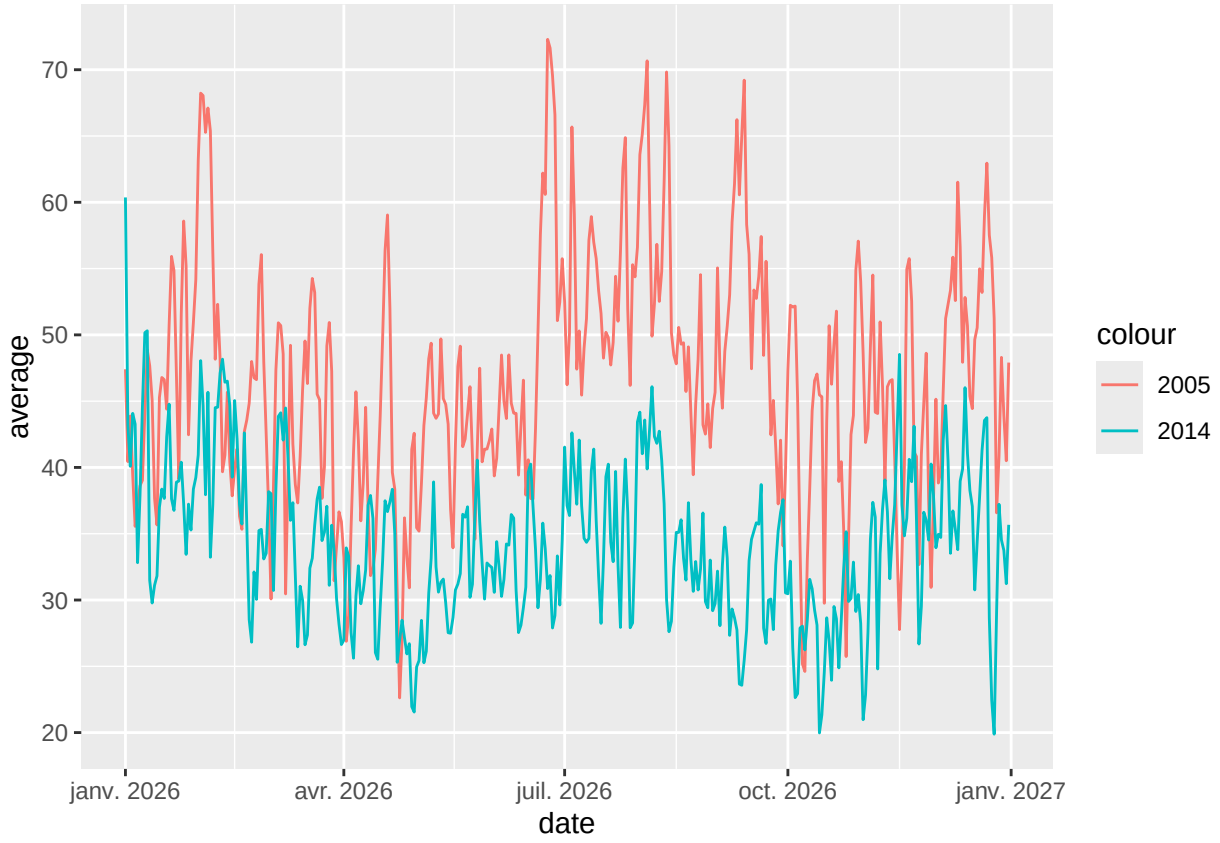

```
plot2 |>
  ggplot() +
  geom_line(aes(x=date, y=average)) +
  # scale_x_log10() +
  labs(title = "average")
```




```
p <- ggplot() +
      # data05 plot
      geom_line(data=plot1, aes(x=date, y=average, color="2005")) +
      # data14 plot
      geom_line(data=plot2, aes(x=date, y=average, color="2014"))



p
```



Question 2

```
# scatter plots by using x-axis
p1 <- data05 |> group_by(Latitude, Longitude)|>
  mutate(location = 1*(AQI>50))

p2 <- data14 |> group_by(Latitude, Longitude)|>
  mutate(location = 1*(AQI>50))


plot1 <- p1 |> ggplot(aes(x=Latitude, y=Longitude)) +
  geom_point(aes(color=factor(location)))

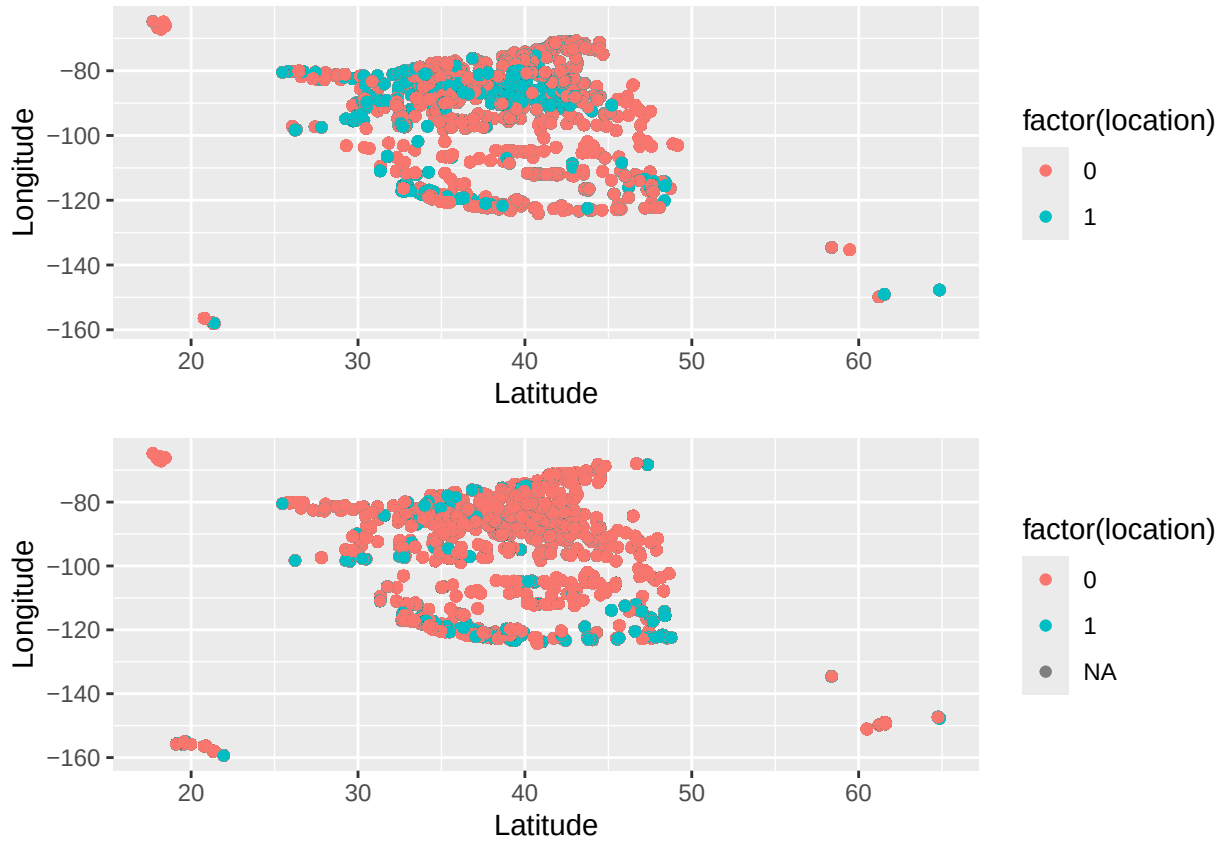
plot2 <- p2 |> ggplot(aes(Latitude, y=Longitude))+
  geom_point(aes(color=factor(location)))

library(gridExtra)
```

```
##
## Attachement du package : 'gridExtra'

## L'objet suivant est masqué depuis 'package:dplyr':
##
##     combine
```

```
grid.arrange(plot1, plot2)
```



Movie Dataset

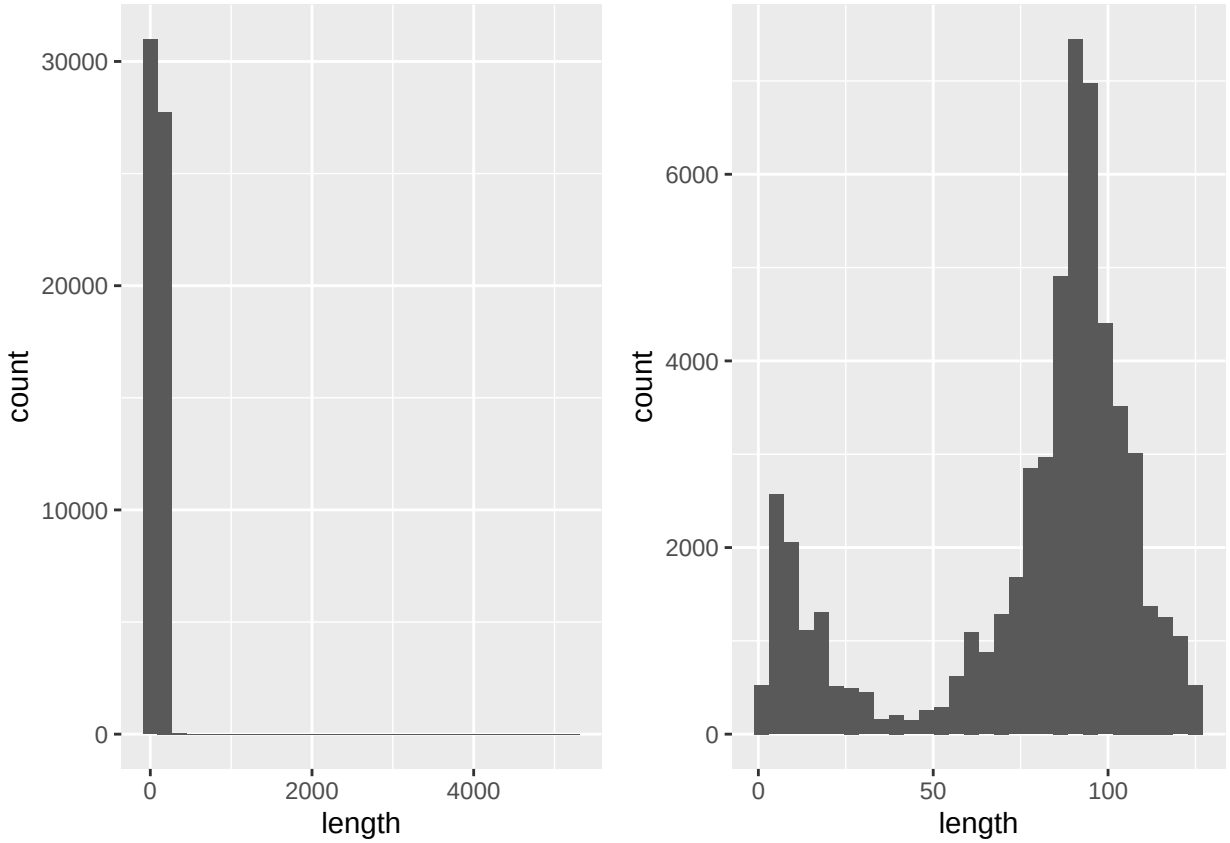
```
library(tidyverse)
movie_data <- read.delim("C:/Users/FRT/Downloads/Fall-2026/MATH-8830/Dataset/movies.tab")

p11 <- movie_data |> ggplot(aes(length)) + geom_histogram() # with outlier


percentile_95 <- quantile(movie_data$length, prob=0.95, na.rm=TRUE)

data1 <- movie_data |> filter(
  !is.na(length),
  length <= percentile_95
)
p12 <- data1 |> ggplot(aes(length)) + geom_histogram(bins=30)
grid.arrange(p11, p12, ncol=2 )
```

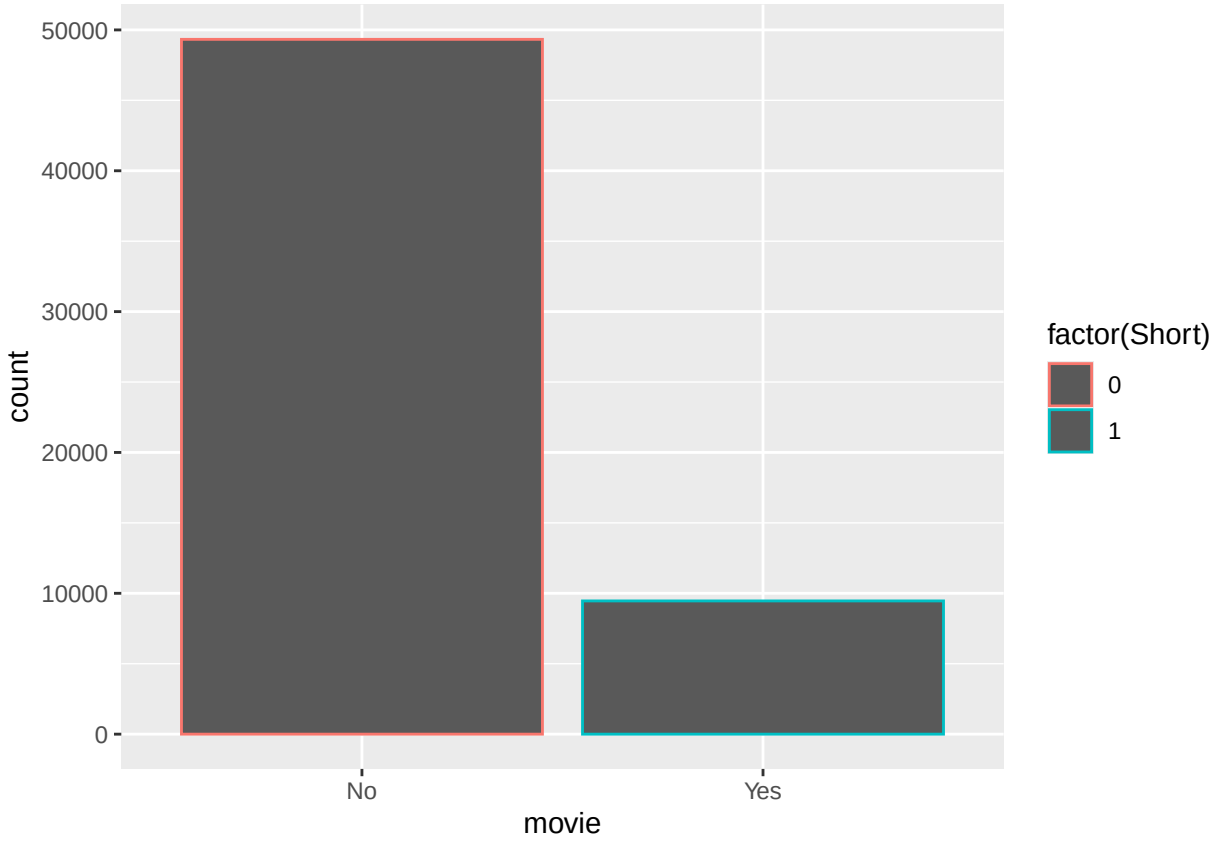
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Question 4

A bar plot to show the frequencies of short movies and the long movies.

```
movie_data |> mutate(
  movie = ifelse(Short==1, "Yes", "No")
)|> ggplot(aes(x=movie, color=factor(Short)))+
  geom_bar()
```



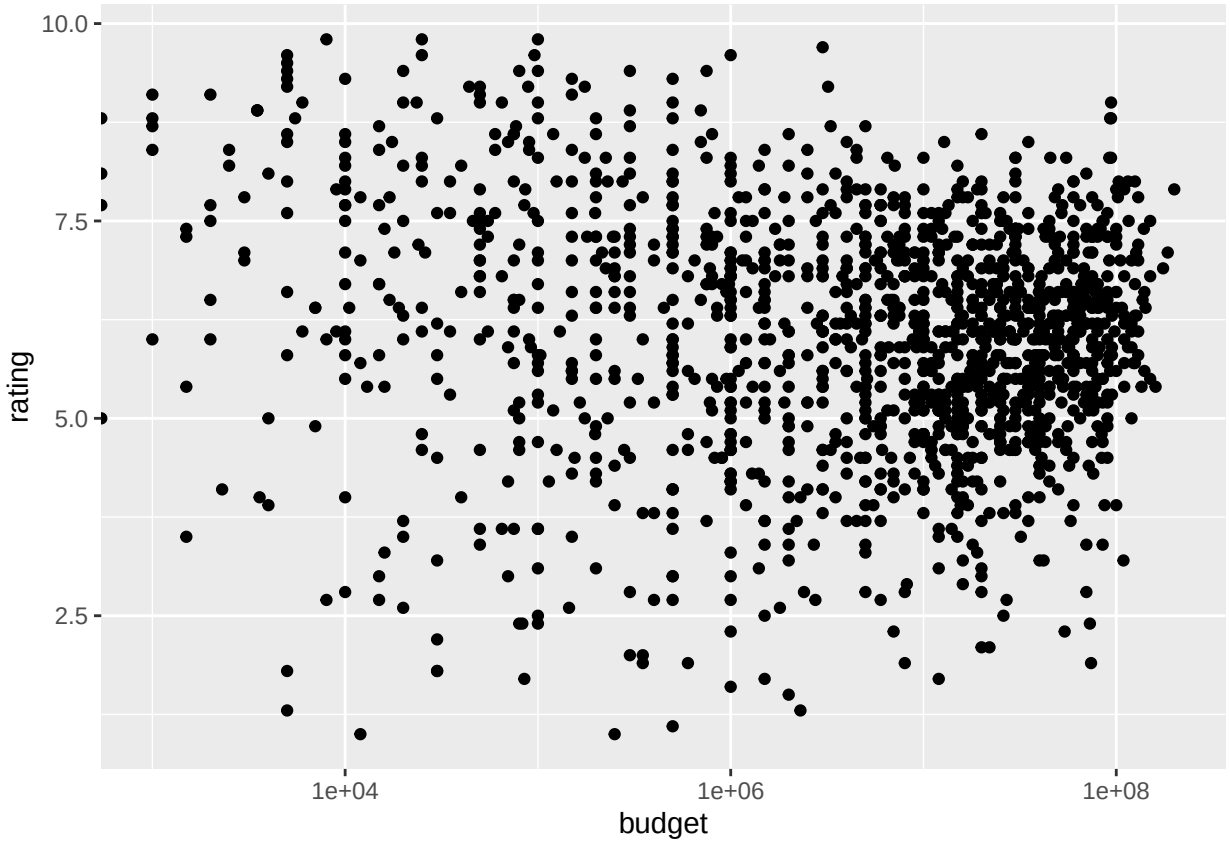
Question 5

```
# relationship between rating and budget

movie_data |> filter(Short==0 & year>=2000) |>
  ggplot(aes(x=budget, y=rating))+
  geom_point()+
  scale_x_log10()
```

```
## Warning in scale_x_log10(): log-10 transformation introduced infinite values.
```

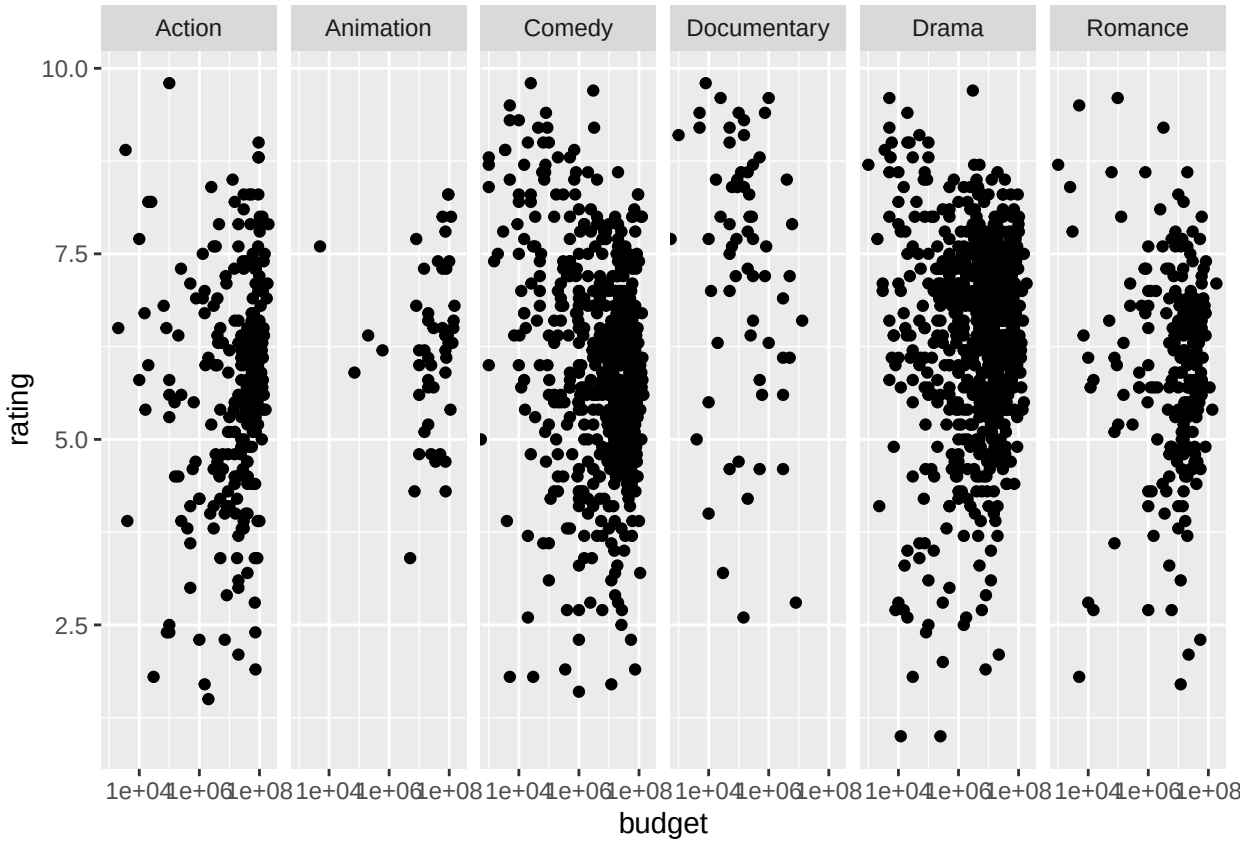
```
## Warning: Removed 6747 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
movie_genre <- c("Action", "Animation", "Comedy", "Drama", "Documentary", "Romance")
movie_data |> filter(Short==0 & year>=2000) |>
  pivot_longer(cols = c("Action", "Animation", "Comedy", "Drama", "Documentary", "Romance"), names_to =
  ggplot(aes(x=budget, y=rating))+
  geom_point()+
  facet_grid(.~genre)+
  scale_x_log10()
```

```
## Warning in scale_x_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 7242 rows containing missing values or values outside the scale range
## (`geom_point()`).
```


Action
Animation
Comedy
Documentary
Drama
Romance
rating
10.0
7.5
5.0
2.5
1e+04
1e+06
1e+08
budget